

Ornithology Week 3 Practice Problems: Phylogenetics and Speciation

- 1) Passeriformes is a “higher” taxon than Psittacidae. (**T, F, or It Depends**)
- 2) Which statement is most accurate?
 - a) Morphological data are better for phylogenetics because they reflect selection
 - b) DNA sequence data are better for phylogenetics because they reflect selection
 - c) Morphological data are better for phylogenetics because they never reflect selection
 - d) DNA sequence data better for phylogenetics because they never reflect selection
 - e) DNA sequence data better for phylogenetics because they provide lots of information
- 3) List reasons why a single species might be “split” into two new species. (1-2 sentences)
- 4) How are the K-Pg boundary and scientific debate about the Hoatzin’s closest relatives related? (1-2 sentences)
- 5) Contrast sympatric, parapatric, and allopatric speciation. Which is most common in birds? (1-2 sentences)
- 6) Having infertile offspring is evidence of prezygotic isolation. (**T, F, or It Depends**)
- 7) Assessing whether an individual bird has a patch of feathers of a specific color is an example of a species delimitation criterion. (**T, F, or It Depends**)